Nested Risk-Aware Preference Distillation via Teacher Value Shaping

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Abstract

Preference alignment has become a central challenge in large language model (LLM) research, yet existing advances remain largely confined to large proprietary models, while small language models (SLMs) continue to suffer from substantial alignment tax due to limited capacity. Recent preference distillation methods mitigate this issue by transferring alignment knowledge from stronger teacher models, but most existing approaches remain risk-neutral and implicitly treat all tokens equally during optimization. In this work, we propose **NARVAL**: **N**ested risk-**A**ware p**R**eference distillation via teacher **VAL**ue shaping, a risk-aware and geometry-aware preference distillation framework for aligning SLMs. NARVAL introduces a nested risk-aware objective based on risk-adjusted teacher value functions to model trajectory-level uncertainty during autoregressive generation, enabling the student model to distinguish between responses with similar expected utility but different risk profiles. In addition, we propose an information-geometric token weighting mechanism that assigns larger optimization weights to alignment-critical tokens according to their influence on the divergence between the aligned teacher model and the reference teacher model. Experiments on diverse preference alignment benchmarks, including reasoning-intensive and safety-critical tasks, demonstrate that NARVAL consistently improves preference distillation performance over existing baselines.

Keywords: Preference Distillation, Teacher Value Shaping, Risk-aware Optimization, Information Geometry

1 Introduction

Preference alignment has become a central objective in large language model (LLM) research, aiming to generate outputs that are not only fluent but also helpful, safe, and consistent with human values. While early approaches relied on Reinforcement Learning from Human Feedback (RLHF) (Christiano et al. 2017), recent methods such as Direct Preference Optimization (DPO) (Rafailov et al. 2023) provide more stable and computationally efficient alternatives. However, these advances have largely been confined to large proprietary LLMs, limiting accessibility and broader adoption due to their high computational cost and closed-source nature (Hsieh et al. 2023). In contrast, small language models (SLMs) offer a cost-effective and efficient alternative for resource-constrained settings, but their limited representational capacity makes effective preference alignment substantially more challenging. Consequently, preference optimization in SLMs often incurs an *alignment tax*, where improved alignment performance degrades general downstream capabilities (Lin et al. 2024).

To improve alignment in small language models (SLMs), recent studies have explored *preference distillation*, where aligned large language models transfer preference knowledge to smaller student models. Unlike conventional knowledge distillation, which primarily matches output distributions (Hinton et al. 2015), preference distillation conveys the teacher's relative judgment of response quality, thereby providing richer supervision for alignment. Representative approaches include distribution-matching and advantage-based methods such as Dual-Constrained Knowledge Distillation (DCKD) and ADPA (Gao et al. 2025), as well as value-guided approaches such as Preference Distillation via Value-Based Reinforcement Learning (TVKD) (Kwon et al. 2025). Although these methods improve the ability of SLMs to distinguish preferred from dispreferred responses and help mitigate the alignment tax (Lin et al. 2024), they remain fundamentally risk-neutral, optimizing only expected reward while overlooking higher-order characteristics of the reward distribution, such as variance and tail risk (Ruszczyński 2010; Chow et al. 2015). Consequently, responses with similar expected utility but substantially different uncertainty profiles may provide nearly indistinguishable supervision, limiting the student's ability to account for potentially unfavorable generation trajectories. Furthermore, most existing methods implicitly treat all tokens equally during the distillation process, despite the fact that different tokens may contribute unequally to the underlying preference signal. Such uniform treatment may dilute the influence of tokens that are particularly important for expressing alignment-relevant behavior. While ADPA introduces implicit action weighting through teacher-derived advantages, its weighting operates over vocabulary-level actions rather than explicitly identifying the importance of individual tokens within a generated sequence.

To address these limitations, we propose **NARVAL**: ***N****ested Risk-****A****ware p****R****eference Distillation via Teacher* ***VAL****ue Shaping*, a risk-aware preference distillation framework that leverages risk-aware teacher value functions to provide uncertainty-sensitive supervision for small language models. Unlike existing preference distillation methods that primarily optimize expected preference utility, NARVAL incorporates a nested risk-aware correction into teacher value estimation, enabling the student model to capture both preference quality and trajectory-level uncertainty

during autoregressive generation. As a result, the student can distinguish between responses with similar expected utility but different risk profiles. Furthermore, we introduce a geometry-aware token weighting mechanism grounded in information geometry (Amari 1998), where token importance is determined by the sensitivity of the divergence between a preference-aligned teacher and a reference teacher model. Tokens associated with larger behavioral divergence are regarded as more alignment-critical and receive higher optimization weights during distillation. By combining sequential risk modeling with geometry-aware token weighting, NARVAL enables the student model to capture uncertainty-sensitive preference structures while improving credit assignment over alignment-critical tokens.

Finally, we evaluate NARVAL on diverse preference alignment benchmarks spanning reasoning-intensive and safety-critical generation tasks. Experimental results and ablation studies demonstrate that both nested risk-aware teacher supervision and geometry-aware token weighting consistently improve preference distillation performance, while providing insights into trajectory-level uncertainty and token-level alignment dynamics in autoregressive preference optimization.

Our main contributions are summarized as follows:

- **Weighted Risk-Aware Preference Optimization.** We propose NARVAL, a geometrically weighted and risk-aware preference distillation framework that incorporates nested risk-aware teacher value shaping to model trajectory-level uncertainty in autoregressive generation.
- **Teacher-guided Geometric Token Importance Weighting.** We introduce a geometry-inspired token weighting mechanism based on the sensitivity of divergence between preference-aligned teacher model and reference teacher model.
- **Empirical Evaluation.** Extensive experiments on reasoning-intensive and safety-critical alignment benchmarks demonstrate that NARVAL consistently improves preference distillation performance across diverse teacher–student settings.

2 Background and Preliminaries

2.1 Preference-based Policy Optimization

We consider a preference dataset consisting of triplets (x, y_w, y_l), where y_w is preferred over y_l for the same prompt x.

Bradley–Terry Model.

To model pairwise preferences, a reward function $r(x, y)$ is commonly trained using the Bradley–Terry model (Bradley and Terry 1952). The probability that y_w is preferred over y_l given x is defined as

$$P_{BT}(y_w \succ y_l|x) = \frac{\exp(r(x, y_w))}{\exp(r(x, y_w)) + \exp(r(x, y_l))} \tag{1}$$

Building upon this, Direct Preference Optimization (DPO) (Rafailov et al. 2023) offers a tractable alternative to reward-model-based Reinforcement Learning from

Human Feedback (RLHF) by directly optimizing the policy from preference data. Rather than learning $r(x, y)$ explicitly, DPO reparameterizes the reward in terms of the log-ratio between the learned policy π_θ and a reference policy π_{ref}, yielding the following loss:

$$\mathcal{L}_{\text{DPO}}(\pi_\theta; \pi_{\text{ref}}) = -\mathbb{E}_{(x,y_w,y_l)\sim\mathcal{D}}\left[\log \sigma\left(\beta \log \frac{\pi_\theta(y_w \mid x)}{\pi_{\text{ref}}(y_w \mid x)} - \beta \log \frac{\pi_\theta(y_l \mid x)}{\pi_{\text{ref}}(y_l \mid x)}\right)\right], \quad (2)$$

where $\sigma(\cdot)$ denotes the sigmoid function and β controls the scale of preference sensitivity. While DPO is computationally efficient and does not require explicit reward modeling, its supervision is inherently coarse-grained: it provides only sequence-level binary feedback, which becomes insufficient for small language models with limited representational capacity (Kwon et al. 2025; Gao et al. 2025).

2.2 Risk Measures

Risk Measures.

In language generation, optimizing solely for expected reward often fails to account for unfavorable outcomes, as such objectives are inherently risk-neutral. To better characterize the distributional behavior of rewards, we employ a risk-sensitive objective (Bäuerle and Rieder 2014; Wang and Chapman 2022) that explicitly captures underlying latent risks.

Nested Risk Measures.

Following Risk-aware Direct Preference Optimization under Nested Risk Measure (Ra-DPO) (Zhang et al. 2025), under Preference-based Markov Decision Process (Pb-MDP), nested risk measures (Fei et al. 2020; Chen et al. 2024; Zhao et al. 2024) can be expressed in a Bellman-style recursive form:

$$\begin{cases} Q_\pi([x, y_{<t}], y_t) = R([x, y_{<t}], y_t) + \Phi_\mu(V_\pi([x, y_{<t+1}])), \\ V_\pi([x, y_{<t}]) = \mathbb{E}_\pi\Big[Q_\pi([x, y_{<t}], y_t)\Big], \\ V_\pi([x, y_{<T}]) = R([x, y_{<T}]). \end{cases} \quad (3)$$

where $\Phi_\mu(\cdot)$ denotes the nested risk measure with risk parameter μ, and Q_π and V_π denote the state–action value and state value under the risk-sensitive criterion.

2.3 Reward Shaping in Reinforcement Learning

Reward shaping has long served as an effective mechanism in reinforcement learning for improving optimization efficiency and mitigating sparse-reward learning (Ng et al. 1999). Recently, this principle has been extended to preference distillation for language models. In particular, TVKD (Kwon et al. 2025) leverages teacher value functions as dense sequential shaping signals for student optimization. However, TVKD remains fundamentally risk-neutral, since the shaping mechanism depends solely on expected future returns estimated by the teacher. As a result, trajectories with similar expected utility but different uncertainty characteristics may receive nearly identical

supervision. In this work, we generalize reward shaping to a risk-aware setting by incorporating nested risk-sensitive teacher value functions, enabling the student model to capture both preference quality and trajectory-level uncertainty during autoregressive generation.

2.4 Information Geometry

Information geometry provides a differential-geometric framework for analyzing parametric families of probability distributions (Amari and Nagaoka 2000; Amari 2016). Let $\mathcal{M} = \{p_\theta : \theta \in \Theta\}$ denote a parametric statistical model parameterized by θ. Under mild regularity assumptions, $\mathcal{M}$ can be viewed as a smooth statistical manifold whose local geometric structure is characterized by the Fisher Information Matrix (FIM) (Rao 1945; Amari and Nagaoka 2000). Intuitively, the FIM defines the intrinsic geometry of the parameter space by measuring the sensitivity of probability distributions to infinitesimal parameter perturbations. Additional details and formulations of the FIM are provided in Appendix A.2.

Local geometry of KL divergence.

A fundamental result in information geometry states that the Kullback–Leibler (KL) divergence between two infinitesimally close distributions admits a second-order Taylor expansion:

$$D_{\mathrm{KL}}(p_\theta \,\|\, p_{\theta+\delta}) = \frac{1}{2}\,\delta^\top F(\theta)\delta + \mathcal{O}(\|\delta\|^3).$$

This approximation suggests that, within a local neighborhood of the parameter space, the Kullback–Leibler (KL) divergence behaves as a squared distance metric on the statistical manifold induced by the Fisher Information Matrix (FIM) (Amari and Nagaoka 2000; Amari 2016). Motivated by this geometric interpretation, we develop a geometric token-importance measure that estimates the contribution of each token to the alignment objective through the local gradient structure of the KL divergence.

3 Methodology

3.1 Teacher-Guided Information-Geometric Token Importance Weight

We propose a token-importance weighting mechanism based on an information-geometric sensitivity analysis of teacher model. Given a prompt x and response sequence $y = (y_1, \ldots, y_T)$, let $\pi_{\mathrm{tch}}^{\mathrm{dpo}}$ denote a preference-aligned teacher obtained via DPO and $\pi_{\mathrm{tch}}^{\mathrm{ref}}$ the corresponding SFT reference model. Since preference alignment shifts the model distribution toward human-preferred responses, the divergence between $\pi_{\mathrm{tch}}^{\mathrm{dpo}}$ and $\pi_{\mathrm{tch}}^{\mathrm{ref}}$ reflects the alignment-induced behavioral transformation. We exploit this divergence signal to estimate token-level importance, assigning larger weights to tokens that contribute more significantly to the distributional shift induced by preference optimization.

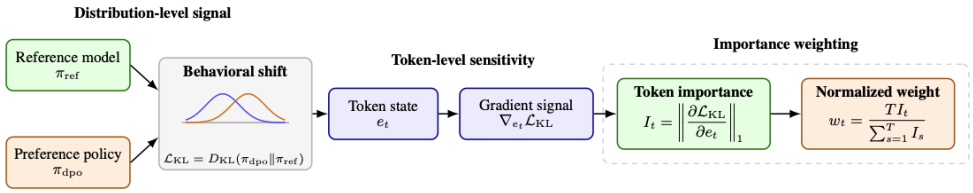


Fig. 1 Overview of the geometric token importance estimation in NARVAL. The behavioral shift between the preference-aligned teacher model $\pi_{\text{tch}}^{\text{dpo}}$ and the reference teacher model $\pi_{\text{tch}}^{\text{ref}}$ induces differences in the generated token distributions.

Statistical manifold of language models.

As discussed in Section 2.4, autoregressive language models can be interpreted as points on a statistical manifold, where the KL divergence between $\pi_{\text{tch}}^{\text{dpo}}$ and $\pi_{\text{tch}}^{\text{ref}}$ approximately reflects the extent to which preference alignment shifts the teacher's model distribution away from the reference teacher model. To quantify the overall behavioral deviation between the two models, we aggregate the per-position divergences:

$$\mathcal{L}_{\text{KL}} = \sum\nolimits_{t=1}^{L} D_{\text{KL}}\Big(\pi_{\text{tch}}^{\text{dpo}}(\cdot \mid \mathbf{s}_{<t}) \,\Big\|\, \pi_{\text{tch}}^{\text{ref}}(\cdot \mid \mathbf{s}_{<t})\Big), \tag{4}$$

where $s_{<t}$ denotes the prefix $[x, y_{<t}]$. This scalar summarizes the cumulative distributional shift introduced by preference alignment across the entire sequence.

Sensitivity with respect to token embeddings.

To identify tokens that contribute most significantly to the divergence between the preference-aligned teacher and the reference teacher, we analyze the sensitivity of the divergence under perturbations to individual token embeddings. Since small embedding perturbations can propagate through the autoregressive generation process and alter the conditional token distributions, the resulting sensitivity provides an information-geometric measure of token importance, which serves as the foundation of our weighting mechanism.

Definition 1 (Geometric Token Importance) The importance of token y_t at position t is defined as the ℓ_1 norm of the gradient of the aggregate KL divergence $\mathcal{L}_{\text{KL}}$ with respect to its embedding $\mathbf{e}_t$:

$$I_t = \left\| \frac{\partial \mathcal{L}_{\text{KL}}}{\partial \mathbf{e}_t} \right\|_1 . \tag{5}$$

Intuitively, I_t measures how strongly perturbations of token y_t affect the divergence between the aligned and reference models. A large value of I_t indicates that small changes in the representation of this token would significantly modify the inter-model divergence, suggesting that the token plays an important role in shaping the alignment-induced behavioral shift. Conversely, tokens with small importance scores contribute little to the overall divergence and thus carry weaker alignment signals.

Importance weight.

We normalize the raw importance scores $\{I_t\}_{t=1}^{T}$ by their sum to eliminate scale sensitivity across batches and training stages:

$$\hat{w}_t = \frac{I_t}{\sum_{s=1}^{T} I_s}. \tag{6}$$

To ensure optimization stability, we rescale the normalized token weights such that their total sum matches the sequence length. This normalization prevents the magnitude of the training objective from fluctuating across sequences of different lengths, thereby ensuring a more stable and length-invariant alignment process. Formally, the final token weight is defined as

$$w_t = T \cdot \hat{w}_t = \frac{T \cdot I_t}{\sum_{s=1}^{T} I_s}, \tag{7}$$

where T denotes the number of response tokens.

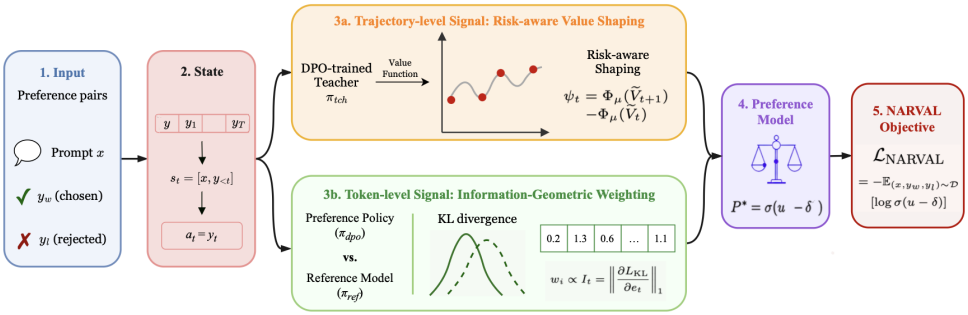


Fig. 2 Overview of NARVAL. Given preference pairs (x, y_w, y_l), NARVAL reformulates generation as a token-level decision process, combines risk-aware teacher value shaping with information-geometric token weighting, and integrates both signals into a preference objective for student distillation.

3.2 Risk-aware Teacher Value Reward Shaping Preference Distillation Loss

State Augmentation.

Although Section 2.2 introduces a recursive Bellman formulation based on nested risk measures, directly applying this framework to language modeling is challenging because preference rewards are only observed at the trajectory level rather than at intermediate token-level states. To overcome this limitation, we adopt the state-augmentation strategy proposed in Ra-DPO (Zhang et al. 2025), reformulating

autoregressive generation as a token-level sequential decision process. Specifically, at timestep t, we define the state as the concatenation of the prompt x and the generated response prefix $y_{<t}$. Under this formulation, the nested risk-aware Bellman equations can be written as follows:

$$\begin{cases} \widetilde{Q}_\pi([x, y_{<t}], y_t) = \Phi_\mu\Big(\widetilde{V}_\pi \ \big(y_{t+1} \circ ([x, y_{<t}], y_t)\big)\Big), \\ \widetilde{V}_\pi([x, y_{<t}]) = \mathbb{E}_\pi\Big[\widetilde{Q}_\pi([x, y_{<t}], y_t)\Big], \\ \widetilde{V}_\pi([x, y_{<T}]) = R([x, y_{<T}]). \end{cases} \tag{8}$$

where $\tilde{Q}^\pi$ and $\tilde{V}^\pi$ denote the risk-aware state–action value and state value functions under policy π, respectively, and $\circ$ denotes sequence concatenation. It is important to note that the computation of $\tilde{V}_\pi([x, y_{<t}])$ differs substantially from that of $V_\pi([x, y_{<t}])$. The detailed relationship will be provided in Appendix B.

Risk-aware Teacher Value Reward Shaping.

Inspired by the key insight of TVKD (Kwon et al. 2025) that teacher value functions provide informative sequential supervision for preference distillation, we propose the risk-aware shaping mechanism by incorporating nested risk-aware teacher value functions. Rather than relying solely on expectation-based value estimates, our formulation explicitly accounts for trajectory-level uncertainty accumulated during autoregressive generation. Concretely, we define a risk-aware shaping potential over the teacher policy π_{tch} using the nested risk-aware value function $\tilde{V}_{\pi_{\text{tch}}}$. The shaping signal is constructed as the temporal difference between consecutive risk-aware teacher potentials:

$$\psi([x, y_{<t}], y_t) = \Phi_\mu\Big(\tilde{V}_{\pi_{\text{tch}}}([x, y_{<t+1}])\Big) - \Phi_\mu\Big(\tilde{V}_{\pi_{\text{tch}}}([x, y_{<t}])\Big), \tag{9}$$

where $\Phi_\mu(\cdot)$ denotes the nested risk operator with risk parameter μ. This shaping signal quantifies how each token affects the teacher's risk-adjusted evaluation of the trajectory. As a result, the student is guided to favor continuations that are not only preference-aligned but also risk-aware, enabling it to distinguish between trajectories with similar expected rewards but different uncertainty profiles.

Risk-aware Preference Distillation Objective.

Building upon the proposed risk-aware teacher value shaping mechanism, we formulate preference distillation as a risk-sensitive optimization problem with a KL constraint toward the teacher policy:

$$\begin{aligned} \max_{\pi_\theta} \ & \mathbb{E}_{y_t \sim \pi_\theta}\Big[r([x, y_{<t}], y_t) + \psi([x, y_{<t}], y_t)\Big] \\ & - \beta \, D_{\text{KL}}\Big(\pi_\theta(\cdot \mid [x, y_{<t}]) \,\|\, \pi_{\text{tch}}(\cdot \mid [x, y_{<t}])\Big) \end{aligned} \tag{10}$$

where $r([x, y_{<t}], y_t)$ denotes the preference reward, $\psi([x, y_{<t}], y_t)$ represents the proposed risk-aware shaping term, and β controls the strength of the distillation constraint

toward the teacher policy. In contrast to conventional preference distillation objectives that optimize only expected preference utility, our formulation explicitly incorporates trajectory-level uncertainty through the nested risk-aware shaping signal. This design encourages the student policy to inherit not only the teacher's preference behavior but also its underlying risk-sensitive structure, leading to more robust and uncertainty-aware alignment.

3.3 Deriving NARVAL Objective via Reformulation of the Bradley–Terry Model

Starting from the constrained optimization problem in Equation 10, we first derive the closed-form optimal solution that maps the risk-aware state–action function $\tilde{Q}_{\pi_{tch}}$ to the optimal policy π^*_θ.

Lemma 1 The constrained problem in Equation 10 admits the following closed-form solution:

$$\pi^*_\theta(y_t \mid [x, y_{<t}]) = \frac{\pi_{\text{tch}}(y_t \mid [x, y_{<t}])}{Z([x, y_{<t}]; \beta)} \exp\left(\frac{1}{\beta}\tilde{Q}_{\pi_{tch}}([x, y_{<t}], y_t)\right), \tag{11}$$

where the partition function:

$$Z([x, y_{<t}];\beta) = \mathbb{E}_{y_t \sim \pi_{\text{tch}}}\left[\exp\left(\frac{1}{\beta}\tilde{Q}_{\pi_{tch}}([x, y_{<t}], y_t)\right)\right]. \tag{12}$$

The detailed derivation of the optimal π^*_θ is provided in Appendix B.2. Then, by rearranging Equation 11, we obtain the expression of the risk-aware state-action function in terms of the policy:

$$\widetilde{Q}_{\pi_{\text{tch}}}([x, y_{<t}], y_t) = \beta \log \frac{\pi^*_\theta(y_t \mid [x, y_{<t}])}{\pi_{\text{tch}}(y_t \mid [x, y_{<t}])} + \beta \log Z(x, y_{<t}; \beta). \tag{13}$$

Although Equation 13 provides a token-level decomposition of the risk-aware utility, it implicitly assumes that all generation steps contribute equally to preference optimization. In practice, however, tokens may exhibit substantially different alignment relevance, since, as discussed in Section 3.1, tokens associated with larger behavioral divergence between the preference-aligned teacher model and the reference teacher model typically encode more informative alignment signals and therefore deserve greater influence during distillation. Motivated by this observation, we introduce an importance-weighted sequential utility estimator that adaptively scales each token-level contribution according to its alignment significance. This weighting mechanism enables finer-grained credit assignment over autoregressive trajectories and allows the student model to focus more strongly on alignment-critical generation steps.

$$\widetilde{Q}_{\pi_{\text{tch}}}([x, y_{<t}], y_t, w^t) = \beta w^t \log \frac{\pi^*_\theta(y_t \mid [x, y_{<t}])}{\pi_{\text{tch}}(y_t \mid [x, y_{<t}])} + \beta w^t \log Z(x, y_{<t}; \beta). \tag{14}$$

In this way, we reformulate the Bradley–Terry model to be directly tied to the optimal policy π^*_θ and the teacher model π_{tch}, as summarized in the following theorem:

Theorem 1 *Given a prompt x and a pair of responses (y_w, y_l), the human preference probability under the Bradley–Terry model can be written as*

$$P^*_{\mathrm{BT}}(y_w \succ y_l \mid x) = \sigma\big(u(x, y_w, y_l, \pi^*_\theta) - \delta(x, y_w, y_l, \pi^*_\theta)\big) \tag{15}$$

where $u(\cdot)$ represents the difference in weighted implicit rewards defined by the student model π_θ and the teacher model π_{tch}:

$$u(x, y_w, y_l, \pi_\theta) = \sum\nolimits_{i=1}^{T_w} w_w^i \beta \log \frac{\pi_\theta(y_w^i \mid x, y_w^{<i})}{\pi_{\mathrm{tch}}(y_w^i \mid x, y_w^{<i})} - \sum\nolimits_{j=1}^{T_l} w_l^j \beta \log \frac{\pi_\theta(y_l^j \mid x, y_l^{<j})}{\pi_{\mathrm{tch}}(y_l^j \mid x, y_l^{<j})}. \tag{16}$$

and $\delta(\cdot)$ is the difference in weighted sequential risk ratios between two pairs (x, y_w) and (x, y_l):

$$\delta(x, y_w, y_l, \pi_\theta) = \beta D_{\mathrm{SeqRR}}(x, y_l; \pi_{\mathrm{tch}} \mid \pi_\theta) - \beta D_{\mathrm{SeqRR}}(x, y_w; \pi_{\mathrm{tch}} \mid \pi_\theta).$$

$$\textit{where } D_{\mathrm{SeqRR}}(x, y; \pi_{\mathrm{tch}} \mid \pi_\theta) = \sum_{t=1}^{T} w^t \Phi_\mu\left(\mathbb{E}_{z \sim \pi_{\mathrm{tch}}}\left[\log \frac{\pi_{\mathrm{tch}}(z \mid [x, y^{<t}])}{\pi_\theta(z \mid [x, y^{<t}])}\right]\right)$$

Our proof is in the Appendix B.3. Therefore, we can obtain the **NARVAL** objective as follows:

$$\mathcal{L}_{\mathrm{NARVAL}} = -\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}}\Big[\log \sigma\big(u(x, y_w, y_l, \pi_\theta) - \delta(x, y_w, y_l, \pi_\theta)\big)\Big] \tag{17}$$

4 Experiments

We evaluate NARVAL on diverse benchmarks. Results show that NARVAL consistently outperforms strong baselines, with ablations confirming the effectiveness of both risk-aware distillation term and proposed weighting mechanism.

4.1 Experimental settings

Backbones and training data.

We evaluate NARVAL across two scales. For the small-scale setting, we use Qwen3 1.7B as the teacher and Qwen3 0.6B as the student. For the large-scale setting, we consider two teacher–student pairs: Llama-3.1 8B → Llama-3.2 1B and Qwen3 8B → Qwen3 1.7B. Both teacher and student models are initialized via supervised fine-tuning (SFT) on Deita10k-V0, which contains approximately 10k high-quality instruction–response pairs. Preference distillation is subsequently performed on UltraFeedback Binarized, a dataset comprising over 63k binarized preference pairs.

Baselines.

We benchmark NARVAL against two baseline categories: preference alignment methods—**DPO** (Rafailov et al. 2023), which directly optimizes the relative log-probability of preferred over rejected responses without an explicit reward model; **WPO** (Zhou et al. 2024), which reweights off-policy preference pairs to better approximate on-policy optimization; and **Ra-DPO** (Zhang et al. 2025), which incorporates risk-aware preference optimization through nested risk measures—and

preference distillation methods—**DCKD** (Gao et al. 2025), which performs contrastive preference distillation to transfer alignment behavior from teacher to student models; **ADPA** (Gao et al. 2025), which adaptively distills alignment preferences via preference-aware objectives; and **TVKD** (Kwon et al. 2025), which transfers token-level preference knowledge through value-guided distillation.

4.2 Main Results on General Capabilities

Table 1 summarizes the general-capability performance across three teacher–student distillation settings. Overall, NARVAL consistently achieves the best average performance among all student alignment methods, demonstrating the effectiveness of the proposed nested risk-aware value shaping strategy and geometric token weighting mechanism. In the Llama-3.1-8B → Llama-3.2-1B setting, NARVAL attains the highest overall average score of 44.21, outperforming strong baselines such as TVKD and ADPA while achieving the best results on HellaSwag, MMLU, and Winogrande. Similar trends are observed in both Qwen settings, where NARVAL obtains the highest average scores of 63.63 and 51.24, respectively, with particularly notable gains on reasoning-intensive benchmarks such as GSM8K and TruthfulQA. Importantly, the improvements remain consistent across different model families and compression ratios, suggesting that the proposed approach generalizes effectively under diverse distillation scenarios.

4.3 Generation Quality

MT Bench.

To further investigate the effect of our framework, we analyze task-wise performance on MT-Bench, as summarized in Table 2. Overall, NARVAL achieves the best average score of 5.21, outperforming all baselines across most categories. In particular, NARVAL shows notable improvements in Writing, Roleplay, Extraction, Math, and Reasoning, suggesting that the proposed value-aware distillation strategy and geometric teacher token weighting effectively transfers both instruction-following and reasoning capabilities from the teacher model. Although TVKD remains competitive in Coding, STEM, and Humanities, NARVAL exhibits more balanced gains across diverse tasks, indicating stronger generalization ability for compact student models.

Harmlessness and Helpfulness.

Table 3 reports pairwise evaluation results on the Anthropic HH-RLHF benchmark, where GPT-5.4-mini is used as the judge model and GPT-4o-mini serves as the reference model following the AlpacaEval protocol. Overall, NARVAL achieves the best performance on both LC win rate and standard win rate, substantially outperforming all baselines. In particular, NARVAL attains a win rate of 14.04, compared to 4.45 for TVKD and 4.52 for ADPA, indicating markedly stronger preference alignment in safety-sensitive dialogue settings. Moreover, the comparable LC win rate between NARVAL and TVKD suggests that the gains are not solely attributable to longer

Method	Hella Swag	ARC	MMLU	Truthful QA	Wino grande	GSM8K	Avg.
Llama-3.1-8B → Llama-3.2-1B							
DPO teacher	83.05	52.30	63.39	57.37	78.22	52.77	64.52
SFT teacher	81.85	51.20	63.05	53.59	78.45	47.84	62.66
SFT	65.84	39.24	31.08	41.13	62.90	7.28	41.25
DPO	66.94	39.25	31.21	44.57	62.12	**7.88**	42.00
WPO	67.44	**40.70**	31.21	42.45	62.27	6.22	41.72
Ra-DPO	66.14	39.40	31.02	41.61	62.83	7.71	41.45
DCKD	65.52	38.29	33.21	48.60	62.27	6.90	42.47
ADPA	65.87	38.57	32.85	**51.46**	62.27	6.88	42.98
TVKD	67.64	39.12	33.00	50.53	62.59	6.91	43.30
NARVAL	**68.20**	40.19	**34.81**	51.35	**62.92**	7.76	**44.21***
Qwen3-8B → Qwen3-1.7B							
DPO teacher	80.60	66.55	77.37	61.52	75.93	80.89	73.81
SFT teacher	79.30	66.04	77.19	58.96	75.69	79.53	72.79
SFT	67.47	52.90	64.90	49.52	64.40	64.37	60.59
DPO	68.44	53.84	64.69	52.54	64.01	65.25	61.46
WPO	68.81	55.29	64.83	53.02	64.01	64.31	61.71
Ra-DPO	67.82	54.10	64.70	56.04	63.93	65.16	61.96
DCKD	66.75	54.92	**65.72**	54.26	64.64	66.49	62.13
ADPA	68.30	**56.23**	64.35	55.50	64.56	64.25	62.20
TVKD	68.47	54.90	63.99	56.52	64.40	64.37	62.61
NARVAL	**69.31**	55.95	64.96	**57.40**	**64.93**	**69.22**	**63.63***
Qwen3-1.7B → Qwen3-0.6B							
DPO teacher	68.44	53.84	64.69	52.54	64.01	65.25	61.46
SFT teacher	67.47	52.90	64.90	49.52	64.40	64.37	60.59
SFT	53.08	37.63	50.59	45.92	59.51	47.84	49.09
DPO	53.74	39.08	50.80	46.92	59.43	52.46	50.41
WPO	52.81	37.80	50.33	47.24	59.51	53.36	50.17
Ra-DPO	52.42	36.24	48.99	45.92	59.43	52.42	49.24
DCKD	53.12	37.84	49.76	46.92	59.27	53.18	50.01
ADPA	**54.10**	39.33	**50.98**	46.11	59.27	53.22	50.50
TVKD	53.42	38.48	50.27	**49.00**	59.98	54.44	50.93
NARVAL	53.85	**39.68**	50.61	48.19	**60.14**	**54.97**	**51.24***

Table 1 General capability results across three teacher–student configurations on HuggingFace Open LLM Leaderboard datasets. The best and second-best student results are shown in **bold** and underline, respectively; * denotes statistical significance with $p < 0.05$.

Method	Writing	Role play	Extraction	Math	Coding	Reasoning	STEM	Humanities	Avg.
SFT	4.30	5.65	1.50	1.20	2.25	1.70	5.25	4.75	3.33
DCKD	7.13	6.85	2.40	2.40	2.30	3.20	5.93	5.70	4.49
ADPA	6.65	6.00	3.30	2.10	2.25	3.05	5.95	5.47	4.35
TVKD	6.70	6.48	2.65	2.15	**3.50**	3.15	**6.58**	**7.65**	4.86
NARVAL	**8.08**	**7.15**	**3.85**	**2.85**	2.35	**3.95**	6.45	6.98	**5.21**

Table 2 Task-wise results for NARVAL and preference distillation benchmarks on MT-Bench.

responses, but instead reflect improved response quality under length-controlled evaluation. These results demonstrate that the proposed framework effectively enhances both helpfulness and harmlessness while mitigating verbosity bias.

Method	LC Win-rate	Win-rate
SFT	0.25	1.46
DCKD	0.85	4.39
ADPA	0.99	4.52
TVKD	3.24	4.45
NARVAL	**3.27**	**14.04**

Table 3 Pairwise evaluation results on LC win-rate and win-rate.

Method	Hella Swag	ARC	MMLU	Truthful QA	Wino grande	GSM8K	Avg.
Llama-3.1-8B → Llama-3.2-1B							
NARVAL without both δ and weight	67.73	38.23	33.87	46.35	62.83	7.13	42.69
NARVAL without δ	68.00	39.59	34.04	44.77	63.06	5.69	42.53
NARVAL without weight in δ	67.71	39.76	33.12	46.30	61.88	5.46	42.37
NARVAL without weight	67.61	39.76	34.14	48.22	61.96	5.47	42.86
NARVAL raw importance weight	65.85	36.14	31.08	47.62	61.14	6.01	41.31
NARVAL normalized weight	67.47	39.55	33.24	49.73	61.58	7.22	43.13
NARVAL ℓ_2-norm weight	68.12	39.74	34.05	50.81	62.11	7.43	43.71
NARVAL	**68.20**	**40.19**	**34.81**	**51.35**	**62.92**	**7.76**	**44.21**
Qwen3-8B → Qwen3-1.7B							
NARVAL without both δ and weight	68.58	54.69	62.61	53.91	64.72	68.69	62.20
NARVAL without δ	68.63	53.75	62.69	53.53	64.17	67.32	61.68
NARVAL without weight in δ	68.30	54.69	62.71	53.95	63.85	67.78	61.88
NARVAL without weight	68.02	53.58	62.70	54.81	64.01	68.54	61.94
NARVAL raw importance weight	68.72	55.02	63.81	55.62	64.38	68.95	62.75
NARVAL normalized weight	68.95	55.31	64.07	56.11	64.55	69.02	63.00
NARVAL ℓ_2-norm weight	69.08	55.58	64.41	56.74	64.71	69.10	63.27
NARVAL	**69.31**	**55.95**	**64.96**	**57.40**	**64.93**	**69.22**	**63.63**

Table 4 Ablation study of NARVAL components across the Llama and Qwen teacher–student configurations.

4.4 Ablation Study and Analysis

We further conduct ablation studies to analyze the contributions of the proposed distillation components, including the risk-aware correction term δ and the geometric token-weighting mechanism. In addition, we investigate the effect of the risk control hyperparameter μ in the proposed objective and provide visualizations of the token-level weighting behavior to better illustrate the effectiveness of the weighting strategy.

To better understand the contribution of each component in NARVAL, we conduct ablation studies on both the Llama and Qwen model families. In particular, we first analyze the contribution of the main components in NARVAL, including the risk-aware correction term δ and the geometric token-weighting mechanism. We then compare several alternative designs for computing token-level weights:

- **Without both δ and weight:** Remove both the risk-aware correction term and the token-level weighting mechanism from the objective.
- **Without δ:** Remove only the risk-aware correction term while retaining geometric token weighting.
- **Without weight in δ:** Remove token-level weighting inside the risk-aware correction term.
- **Without weight:** Remove the geometric token-weighting mechanism entirely and use uniform token contributions.
- **Raw importance:** Directly use the token importance scores I_t as weights without normalization or sequence-length scaling.
- **Normalize:** Normalize the token importance scores across the sequence, but omit the sequence-length scaling factor used in the full method.
- **L2 norm:** Compute token importance using the ℓ_2 norm of the gradient instead of the ℓ_1 norm used in the proposed method.

Table 4 presents the ablation results of different components and weighting strategies in NARVAL across the Llama and Qwen model families. Overall, the full framework consistently achieves the best performance on all benchmarks, demonstrating the effectiveness of combining risk-aware correction with geometric token-level weighting. Removing either the risk-aware term δ or the weighting mechanism leads to noticeable performance degradation, while removing both components results in the largest drop, highlighting their complementary contributions to preference distillation. We further compare several alternative weighting strategies, including raw importance weighting, normalized weighting, and ℓ_2-norm-based weighting. Although these variants improve over simpler ablations, they remain consistently inferior to the proposed geometric weighting formulation. In particular, normalized and ℓ_2-based variants yield relatively stronger performance, suggesting that stable and calibrated token importance estimation plays an important role in effective preference transfer. These results collectively demonstrate that both the risk-aware objective and the proposed weighting design are crucial for achieving robust alignment performance.

Qualitative Analysis of Geometric Weights.

Figure 3 visualizes the token-level importance weights assigned by the proposed teacher-guided geometric token weighting mechanism during preference distillation. The visualization shows that the model consistently assigns higher weights to semantically informative and task-relevant tokens (e.g., command keywords, file extensions, and critical operations), while less informative connective or auxiliary tokens receive comparatively lower weights. This behavior demonstrates that the proposed weighting strategy can effectively identify the most preference-relevant tokens from teacher responses, enabling the student model to focus on critical decision-making signals during optimization. Such selective emphasis improves the quality of preference transfer and provides empirical evidence that the geometric weighting mechanism captures meaningful token-level importance patterns rather than distributing weights uniformly across the sequence.



Fig. 3 Visualization of geometric token importance weights. Token-level weights assigned to the preferred and rejected responses for a shell-command example. Darker shading indicates greater importance under the proposed teacher-guided geometric weighting mechanism.

Analysis of μ on performance of NARVAL.

Table 5 presents the sensitivity analysis of NARVAL under different values of the risk parameter μ. Overall, the performance remains consistently stable across all configurations, with only marginal fluctuations in the average score. Specifically, the average performance varies within a narrow range from 62.97 to 63.63, indicating that NARVAL is not overly sensitive to the choice of μ. In this work, we adopt $\mu = 0.97$ as the default setting since it achieves the best overall average performance while providing balanced improvements across most benchmarks. These results suggest that the proposed framework maintains robust behavior under different risk parameter settings, demonstrating strong stability with respect to the hyperparameter μ.

Method	Hella Swag	ARC	MMLU	Truthful QA	Wino grande	GSM8K	Avg.
NARVAL ($\mu = 0.99$)	68.92	55.48	64.31	56.72	64.55	68.84	63.14
NARVAL ($\mu = 0.98$)	69.08	55.71	64.67	57.08	64.76	69.01	63.39
NARVAL ($\mu = 0.97$)	69.31	55.95	64.96	57.40	64.93	69.22	63.63
NARVAL ($\mu = 0.95$)	68.84	55.22	64.18	56.54	64.41	68.63	62.97

Table 5 Sensitivity analysis of NARVAL under different risk parameters μ.

5 Conclusion

We introduced NARVAL, a preference distillation framework that combines nested risk-aware value shaping with geometric token-level weighting for compact language models. By incorporating trajectory-level uncertainty and token importance into the distillation objective, NARVAL provides richer alignment supervision than existing risk-neutral approaches. Experiments across multiple teacher–student settings demonstrate consistent improvements over strong baselines on general capability, generation quality, and safety-oriented benchmarks. These results highlight the effectiveness of risk-aware preference distillation for improving the alignment and generalization ability of small language models.

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- **Competing Interests:** The authors declare that they have no competing interests.
- **Availability of data and material:** The datasets used in this study are publicly available. Their sources and preprocessing procedures are described in the experimental section.
- **Code availability:** The implementation for NARVAL can be found in https://github.com/PhiVanDat123/NARVAL.
- **Authors' contributions:** All authors contributed to the study conception and design. Material preparation, data collection and analysis were performed by **Dat Phi Van** and **Ngoc Anh Pham**. The first draft of the manuscript was written by **Dat Phi Van** and **Ngoc Anh Pham** and they also analyzed the theoretical properties and developed new theories. All authors revised the final manuscript.
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- **Consent to participate:** Not applicable
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Appendix A Supplementary Materials for Section 2

A.1 Conditional Value-at-Risk (CVaR).

For a risk level $\alpha \in (0, 1)$, the CVaR is defined as:

$$\rho^{\pi}_{\mathrm{CVaR}}(G; \alpha) = \min_{\eta \in \mathbb{R}} \left\{ \eta + \frac{1}{1-\alpha} \mathbb{E}^{\pi} \left[(G - \eta)_+ \right] \right\}, \tag{A1}$$

where $(x)_+ = \max(x, 0)$. CVaR captures tail risks and is a coherent risk measure.

In this paper, we formulate nested risk measures within the Pb-MDP framework and express them in a Bellman-style recursive form. For simplicity, we denote such nested risk measures by Φ_μ, where μ is a risk control parameter.

A.2 Fisher information.

The Fisher Information Matrix at parameter θ is defined as

$$F(\theta) = \mathbb{E}_{x \sim p_\theta} \left[\nabla_\theta \log p_\theta(x) \nabla_\theta \log p_\theta(x)^\top \right]. \tag{A2}$$

The FIM induces a natural Riemannian metric on the statistical manifold and plays a central role in characterizing the local sensitivity of probabilistic models (Amari and Nagaoka 2000; Cover and Thomas 2006).

Appendix B Supplementary Materials for Section 3

B.1 Relationship to Standard Value Functions

Let

$$R_{1:t-1} = \sum_{h=1}^{t-1} \gamma^{h-1} R([x, y_{<h}], y_h) \tag{B1}$$

denote the cumulative reward collected before timestep t. The standard and augmented risk-aware value functions satisfy

$$\widetilde{V}_\pi([x, y_{<t}]) = V_\pi([x, y_{<t}]) + R_{1:t-1}. \tag{B2}$$

Proof.

The result follows by backward induction. At the terminal step, it follows directly from the trajectory-reward decomposition. Assume that Equation B2 holds at timestep $t + 1$. Substituting the induction hypothesis into the augmented Bellman recursion and applying the translation invariance of Φ_μ gives

$$\widetilde{V}_\pi([x, y_{<t}]) = V_\pi([x, y_{<t}]) + R_{1:t-1}. \tag{B3}$$

Therefore, the relation holds for every timestep. At $t = 1$, we obtain $\widetilde{V}_\pi([x]) = V_\pi([x])$.

B.2 Proof of Lemma 1

Proof.

For a fixed prefix $[x, y_{<t}]$, define

$$Z([x, y_{<t}]; \beta) = \mathbb{E}_{y_t \sim \pi_{\text{tch}}} \left[\exp\left(\frac{\widetilde{Q}_{\pi_{\text{tch}}}([x, y_{<t}], y_t)}{\beta} \right) \right]. \tag{B4}$$

Using this partition function, the objective in Equation 10 can be rewritten as

$$\begin{aligned} \max_{\pi_\theta} \; & -\beta D_{\text{KL}}\left(\pi_\theta(\cdot \mid [x, y_{<t}]) \,\middle\|\, \frac{\pi_{\text{tch}}(\cdot \mid [x, y_{<t}])}{Z([x, y_{<t}]; \beta)} \exp\left(\frac{\widetilde{Q}_{\pi_{\text{tch}}}([x, y_{<t}], \cdot)}{\beta} \right) \right) \\ & - \Phi_\mu\left(\widetilde{V}_{\pi_{\text{tch}}}([x, y_{<t}]) \right) + \beta \log Z([x, y_{<t}]; \beta). \end{aligned} \tag{B5}$$

The last two terms are independent of π_θ. Since the KL divergence is non-negative, the objective is maximized when it is zero. Therefore,

$$\pi^*_\theta(y_t \mid [x, y_{<t}]) = \frac{\pi_{\text{tch}}(y_t \mid [x, y_{<t}])}{Z([x, y_{<t}]; \beta)} \exp\left(\frac{\widetilde{Q}_{\pi_{\text{tch}}}([x, y_{<t}], y_t)}{\beta} \right), \tag{B6}$$

which proves Lemma 1.

B.3 Proof of Theorem 1

Proof.

From the weighted policy representation in Equation 14, the risk-aware advantage accumulated along a trajectory y can be written as

$$\sum_{t=1}^{T} \widetilde{A}_{\pi_{\text{tch}}}([x, y_{<t}], y_t) = \beta \sum_{t=1}^{T} w_t \log \frac{\pi^*_\theta(y_t \mid [x, y_{<t}])}{\pi_{\text{tch}}(y_t \mid [x, y_{<t}])} + \beta D_{\text{SeqRR}}(x, y; \pi_{\text{tch}} \mid \pi^*_\theta), \tag{B7}$$

where translation invariance of Φ_μ removes the state-dependent partition terms. Following the token-level language-generation setting, we set $\gamma = 1$.

Taking the difference between the preferred trajectory y_w and the rejected trajectory y_l in Equation B7 gives

$$u(x, y_w, y_l, \pi^*_\theta) - \delta(x, y_w, y_l, \pi^*_\theta), \tag{B8}$$

where u and δ are defined in Theorem 1. Substituting this difference into the Bradley–Terry model yields

$$P^*_{\text{BT}}(y_w \succ y_l \mid x) = \sigma\left(u(x, y_w, y_l, \pi^*_\theta) - \delta(x, y_w, y_l, \pi^*_\theta) \right). \tag{B9}$$